

UNIVERSITY OF TECHNOLOGY - VNUHCM	HOMEWORK 4	Semester/Academic year	1 / 2026-2027
FACULTY OF COMPUTER SCIENCE AND ENGINEERING	Probabilistic Learning and Graphical Reasoning	Due	12/10/2026 23:59
	CO3117 - Machine Learning	Format	Formative take-home

Purpose. Strengthen probabilistic reasoning, Naive Bayes, evidence handling, and graphical-model intuition while avoiding direct rehearsal of midterm formulations.

- Complete the work by hand where calculation or derivation is requested. Show reasoning; unsupported final numbers receive limited credit.
- Use lecture notes and course resources. Do not use AI, online solution lookup, or classmates' answers for your first attempt. Later supportive AI use must follow the course AI_USE.md protocol.
- These activities are designed for concept transfer and assignment support. They intentionally do not reproduce the wording, ordering, numerical data, or question structure of the midterm.
- No official worked solutions will be released. Students are encouraged to self-check their handwriting with course materials, discussion, and AI assistants after completion.
- Submit one PDF in your course repository and preserve your first attempt. Corrections may be added later without replacing the original submission.

Activity 1 - Likelihood as evidence (1.5 pts)

A biased coin produces 9 heads in 12 tosses. Write the likelihood function up to proportionality and explain how the observed count shapes which values of the head probability are plausible. You do not need to derive an estimator.

Activity 2 - Naive Bayes by hand (2.5 pts)

- A small training set uses two binary attributes A and B and class Y in {0,1}: (0,0,0), (0,1,0), (1,0,1), (1,1,1), (1,0,1), (0,1,1), (0,0,0), (1,1,0). Using add-one smoothing, classify the new case A=1,B=0. Show every probability used.
- After the calculation, identify one conditional-independence assumption that may be unrealistic in real data.

Activity 3 - Read a Bayesian-network graph (2.0 pts)

Consider the DAG $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow D$, $B \rightarrow E$. For each pair, state whether a path is blocked or active under the given evidence and justify using d-separation ideas: (i) A and B with no evidence; (ii) A and B given C; (iii) A and D given C; (iv) C and E given B.

Activity 4 - Evidence changes beliefs (1.5 pts)

- Give one concrete everyday example of predictive reasoning and one of diagnostic reasoning. For each, identify what is treated as cause/evidence and what is inferred.
- Explain in one paragraph why observing a common effect can make two previously independent causes statistically dependent.

Activity 5 - Assignment bridge: probabilistic model check (2.5 pts)

- Use a probabilistic model relevant to your longitudinal-assignment data (e.g., Naive Bayes or another approved model). Report the representation used, the assumptions you are making, one calibration/probability diagnostic if feasible, and one failure case caused by a violated assumption.
- Write a 5-8 sentence model-suitability note suitable for your MODEL_LOG. Focus on evidence from your own data rather than textbook claims.

Submission checklist

- ☐ Name and student ID on first page
- ☐ All calculations show intermediate steps
- ☐ Assignment-bridge activity uses your own dataset/use case
- ☐ First attempt preserved
- ☐ Any later AI/reference use disclosed according to course policy